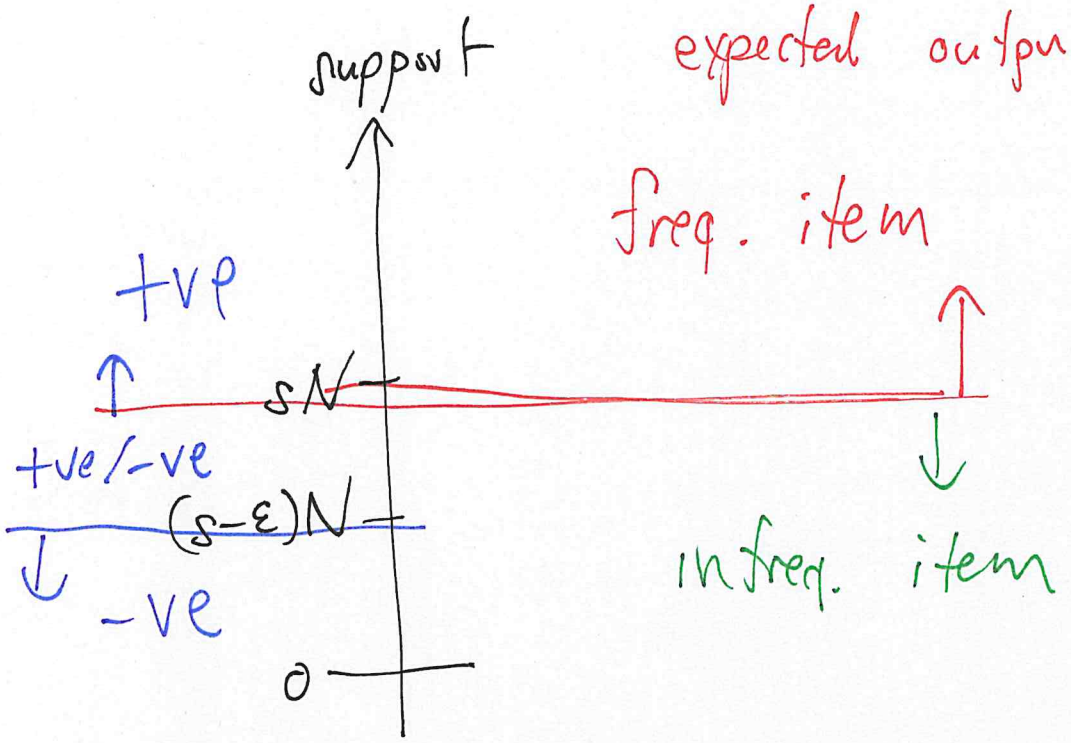
Ex. 8

Q2

	b First Choice	f Second Choice	d Third Choice
b	(250)		
c	125	50	25
d	160	60	(60)
e	210	60	50
f	120	(70)	
g	99	49	49
h	90	40	30

Resulting views = { b, f, d }


support
expected output
freq. item
+ve
sN
+ve/-ve
(s-ε)N
-ve
infreq. item
0

Ex 9

Q1

	false +ve	false -ve
Alg. 1	/	I_2
Alg. 2	I_3	/
Alg 3	I_3	I_2

Q2 $\varepsilon N = 2$

$(s-\varepsilon)N = 2$

Alg 1

- ~~has~~ / does NOT have an ε-deficient synopsis

Cond. 2 is violated — The diff. bet. the est. freq. and the true freq. of $I_1 = 7-4 = 3$ which is ~~at most~~ / greater than εN

Alg. 2

- has / ~~does NOT have~~ an ε-deficient synopsis

Cond. 1 — no false -ve

Cond 2 — the difference bet. the est. freq. and the true freq of each item $\leq \varepsilon N$ (i.e. the diff. for I_1, I_2, I_3 and I_4 are 2, 1, 1, 0, respectively)

Cond 3 — All items whose true freq. less than $(s-\varepsilon)N$ (i.e, I_4) are classified as infreq. item in the alg. output

Alg. 3

- ~~has~~ / does NOT have an ε-deficient synopsis

Cond. 3 is violated — The item whose true freq. less than $(s-\varepsilon)N$ (i.e., I_4) ~~is~~ / is NOT as an infreq. item in the alg. output

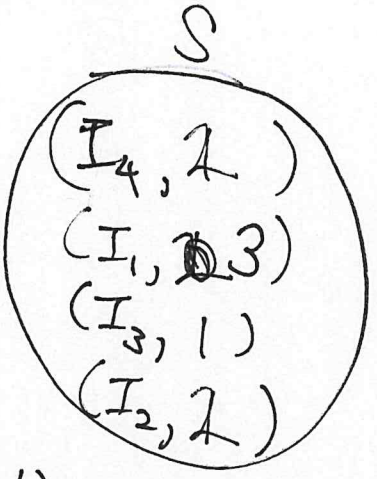
- probabilistic approach

- deterministic approach

Q3(a) $t = 4$

(b)(i) $x = 8$

(ii)



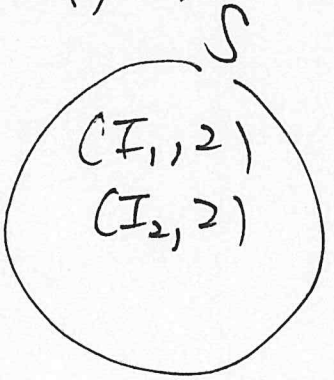
(iii)(1) $r_1 = 2$

(2) $(I_1, 3) \xrightarrow{T} (I_1, 2) \xrightarrow{H} (I_1, 2)$

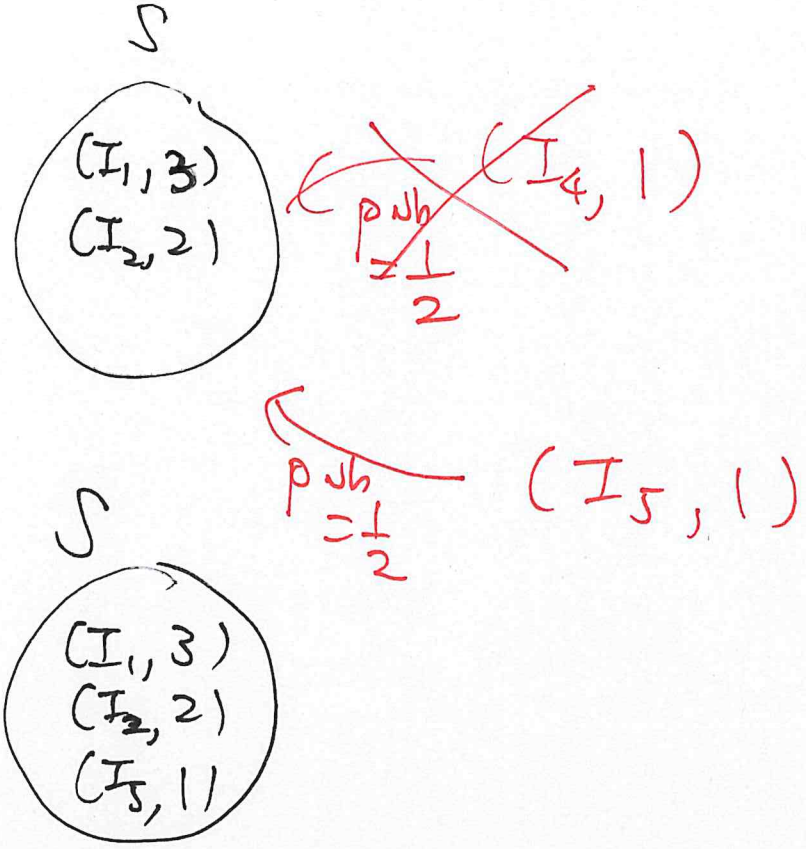
$(I_2, 2) \xrightarrow{H} (I_2, 2)$

$(I_3, 1) \xrightarrow{T} (I_3, 0)$ ← delete entry

$(I_4, 2) \xrightarrow{T} (I_4, 1) \xrightarrow{T} (I_4, 0)$ ↳ delete entry



Q3(b)(iv)



(c)(i) $y = 16$

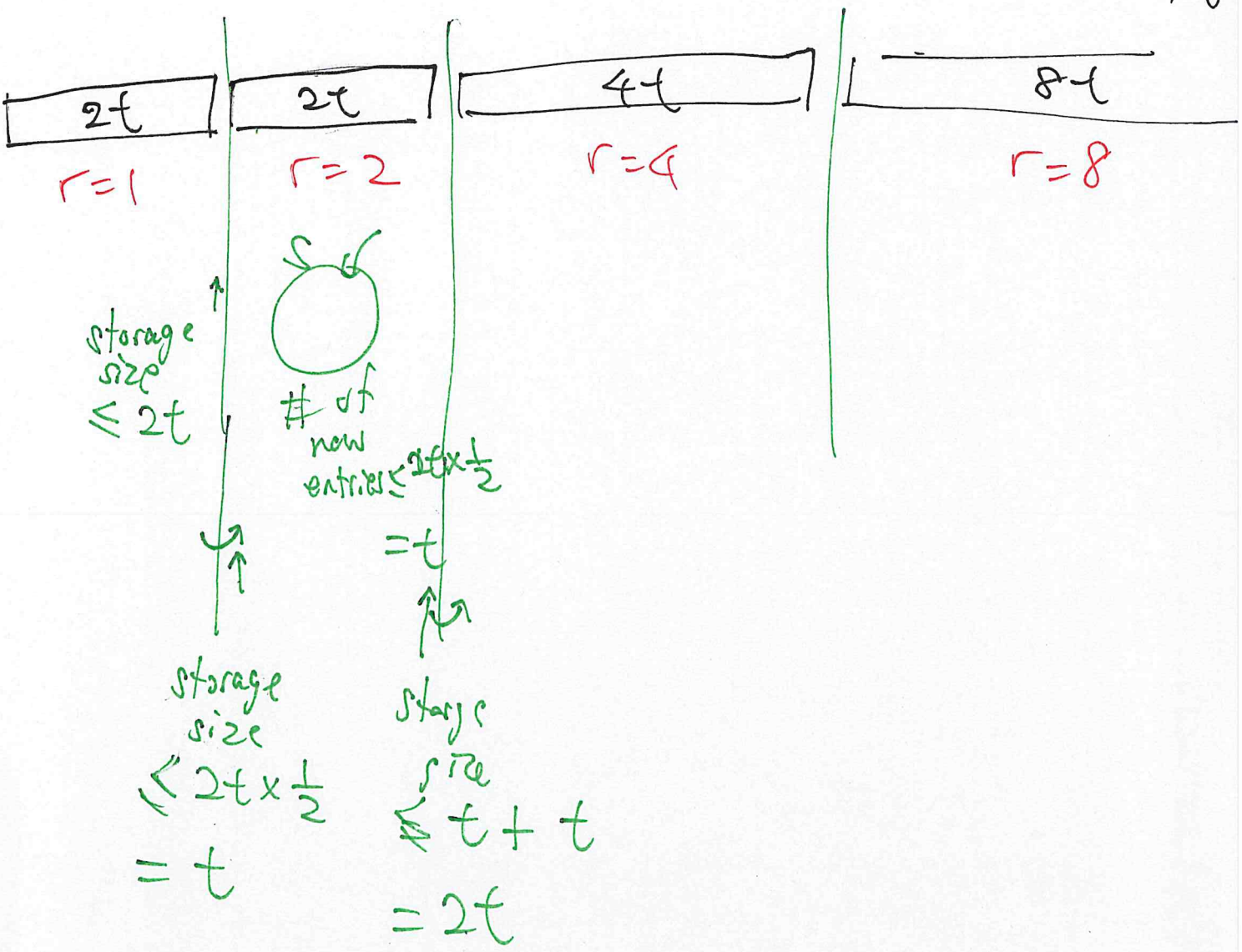
(ii) $r_2 = 4$

(iii) $\varepsilon N = 0.35 \times 16 = 5.6$

$sN = 0.5 \times 16 = 8$

	f	$f + \varepsilon N$	$f + \varepsilon N \geq sN$?
I_1	5	10.6	Yes
I_2	4	9.6	Yes
I_3	2	7.6	No

output = $\{I_1, I_2\}$


2t
2t
4t
8t
r=1
r=2
r=4
r=8
storage size ≤ 2t
of new entries ≤ 2t x 1/2 = t
storage size ≤ 2t x 1/2 = t
storage size ≤ t + t = 2t